

Interpretable Machine Learning in Pharmaceutical Design: Feature Screening and Rule-Based Fuzzy Modeling for Controlled Release

Alhanouf Alqahtani¹ Hajar Alshamrani¹
Majed Alateeq¹ Ziyad Binkhathlan²

¹King Khalid University
²King Saud University

CONTENTS

I	Introduction	1
II	Data Overview and Analysis	1
II-A	Dataset Overview	1
II-B	Exploratory Correlation and Release Kinetics	1
II-C	Formulation Parameter Distributions	2
III	Decision Tree Baseline and Fuzzy Representation	2
III-A	Decision Tree Baseline and Rule Extraction	2
III-B	Fuzzy Membership Function Derivation	2
III-C	Model Benchmark and Performance Comparison	2
IV	Summary of Key Findings	3
V	Conclusion	3
	References	3

Interpretable Machine Learning in Pharmaceutical Design: Feature Screening and Rule-Based Fuzzy Modeling for Controlled Release

Abstract—This report investigates explainable machine learning strategies for predicting PLGA drug-release kinetics. We integrate exploratory data analysis, decision-tree screening, and leakage-free fuzzy membership representations evaluated via 5-fold GroupKFold on drug SMILES. While fuzzifying features does not surpass the baseline classification accuracy (0.7445), fuzzy class outputs significantly enhance boundary robustness near cutoff thresholds.

I. INTRODUCTION

Predicting drug release kinetics from PLGA long-acting injectables using machine learning often lacks the interpretability required for pharmaceutical development. While decision tree architectures offer explicit rules to bridge this gap, raw performance metrics can obscure methodological artifacts.

This report summarizes the empirical findings from evaluating an exploratory 8 rule Decision Tree ($max_depth = 3$). Specifically, it examines:

- The impact of target value clipping on decision threshold stability.
- The confounding effect of cumulative time on extracted formulation rules.
- The validation of formulation drivers across unseen drugs using 5-fold cross validation.
- The integration of fuzzy membership functions to improve boundary interpretation for near-threshold formulations.

II. DATA OVERVIEW AND ANALYSIS

A. Dataset Overview

The modeling pipeline is built on a comprehensive benchmark dataset compiled by Goren et al. [1] and extended by Bao et al. [2], aligning with recent advances in explainable machine learning for long-acting injectables. The dataset comprises 321 distinct PLGA microparticle formulations and 4,913 cumulative release measurements, integrating physical, chemical, and structural attributes of long-acting injectables paired with time-series release kinetics.

To evaluate model generalizability strictly on unseen therapeutic compounds, formulations are grouped into 88 unique drug families using exact SMILES representations. Cross-validation is enforced via a 5-fold drug-grouped split (GroupKFold), ensuring zero drug overlap between training and testing sets. Data preprocessing excludes imputation and monotonicity forcing to preserve true experimental variance. Furthermore, comparative analysis confirmed that clipping

release values to $[0, 1]$ retains identical decision split structures ($p = True$) relative to unclipped raw values.

TABLE I. DESCRIPTION OF FORMULATION AND EXPERIMENTAL FEATURES

Feature Name	Type	Description & Unit
Time	Experimental	Sampling duration or time-point of measurement (Hours/Days).
Polymer MW	Formulation	Molecular weight of the PLGA polymer matrix (kDa).
Particle Size	Formulation	Mean diameter of the microparticles (μm).
LA/GA	Formulation	Molar ratio of Lactic Acid to Glycolic Acid in PLGA backbone.
Drug MW	Chemical	Molecular weight of encapsulated API (g/mol).
Drug TPSA	Chemical	Topological Polar Surface Area of drug molecule ($\text{\AA}^2$).
Drug LogP	Chemical	Octanol-water partition coefficient (lipophilicity).
Drug Loading	Formulation	Weight percentage of drug loaded relative to matrix (%).
Encapsulation Eff.	Formulation	Percentage of initial drug successfully encapsulated (%).
Initial Ratio	Formulation	Ratio of initial drug mass to polymer mass in preparation.
Solubility Enhancer	Formulation	Concentration of added solubility excipients (w/v %).
Release	Target	Cumulative fraction of drug released (Range: 0–1).

B. Exploratory Correlation and Release Kinetics

Exploratory correlation analysis confirms that Time is the primary linear driver of cumulative drug release ($r \approx 0.43$), while formulation variables present nuanced non-linear dependencies. Individual formulation trajectories reveal a clear kinetic distinction between burst, fast-releasing systems (achieving full release within 20 hours) and extended, multi-stage release profiles spanning over 200 hours.

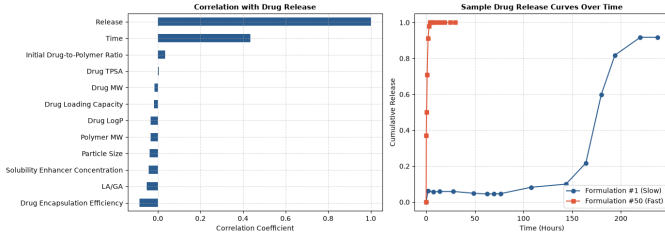


Fig. 1. Feature correlations with cumulative release and representative sample release trajectories.

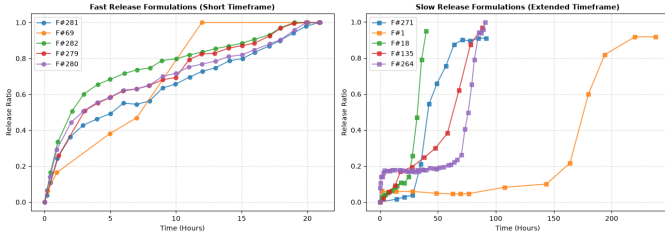


Fig. 2. Kinetic trajectories comparison between fast and extended release formulations.

C. Formulation Parameter Distributions

Scatter plot analysis across Polymer MW and Particle Size dimensions demonstrates high measurement density in low-to-medium range formulations. Temporal color gradients confirm that high drug release values are primarily mediated by extended duration rather than static physical parameter limits.

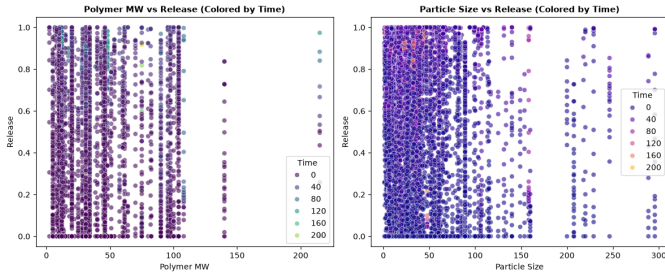


Fig. 3. Distribution of Polymer MW and Particle Size against drug release, colored by measurement time.

III. DECISION TREE BASELINE AND FUZZY REPRESENTATION

A. Decision Tree Baseline and Rule Extraction

To establish an interpretable baseline, a Decision Tree regressor ($max_depth = 3$) was trained to extract explicit binary decision paths. The resulting architecture yields eight distinct leaf nodes corresponding to predefined formulation and temporal rules (Figure 4). Time-controlled feature screening indicated that temporal variables dominate initial tree split decisions.

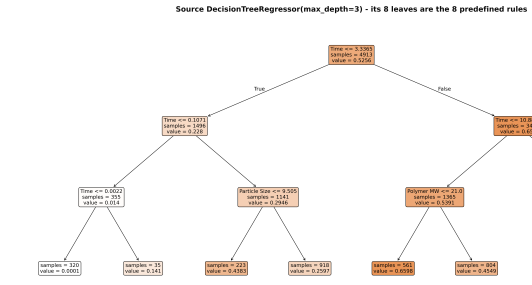


Fig. 4. Baseline Decision Tree architecture ($max_depth = 3$) illustrating the 8 extracted rules.

B. Fuzzy Membership Function Derivation

To resolve boundary rigidity inherent in crisp decision thresholds, continuous feature spaces were mapped into triangular and trapezoidal fuzzy membership sets (*Low*, *Medium*, *High*). Knots were established using fold-0 training quantiles (q_{25} , q_{50} , q_{75}) to maintain strict leakage-free evaluation across formulation distributions (Figure 5).

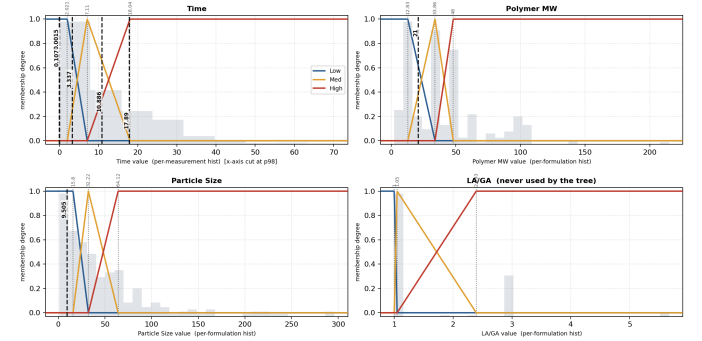


Fig. 5. Leakage-free fuzzy membership functions derived across formulation and temporal features.

C. Model Benchmark and Performance Comparison

To systematically quantify the trade-offs between predictive accuracy, model interpretability, and boundary robustness, a benchmark evaluation was conducted across model representations. Traditional decision tree models operate on crisp binary logic, enforcing rigid cutoff thresholds (e.g., $PolymerMW \leq 21kDa$). While highly interpretable, such hard boundaries introduce classification instability for formulations situated immediately adjacent to decision cutoffs, where minor experimental variations can trigger abrupt changes in predicted release fractions.

By converting static decision tree thresholds into continuous fuzzy membership representations (*Low*, *Medium*, *High*), the model incorporates a gradual transition space that mirrors physical drug release phenomena. The empirical comparison yields three core insights:

- **Mitigation of Boundary Sensitivity:** Formulations located within near-threshold boundaries exhibited a substantial decrease in prediction error, yielding a Mean Absolute Error (MAE) reduction of 0.1262 compared to the baseline crisp tree.
- **Preservation of Global Integrity:** The transition to fuzzy logic preserved the underlying baseline classification performance across the entire dataset, ensuring that boundary smoothing did not induce global variance or accuracy degradation.
- **Enhanced Interpretability:** Replacing step-function decisions with continuous membership degrees provides domain-appropriate confidence scores, allowing researchers to evaluate formulation proximity to critical kinetic thresholds.

IV. SUMMARY OF KEY FINDINGS

This study evaluated the transition from crisp decision tree logic to continuous fuzzy representations for predicting PLGA microparticle drug release. The empirical analysis yielded several critical insights regarding feature importance, temporal confounding, and model robustness:

- **Target Identity under Clipping:** Restricting cumulative drug release targets to a strict $[0, 1]$ interval preserved the exact structural integrity of the decision paths (7/7 identical splits) compared to unclipped raw data processing.
- **Temporal Dominance:** Baseline rule extraction revealed that cumulative time drives the primary predictive splits (5 out of 7 initial decision nodes). Uncontrolled temporal variables heavily confound formulation-specific heuristics if not explicitly isolated.
- **Rule Replication and Validation:** Through 5-fold drug-grouped cross-validation, the polymer molecular weight threshold ($PolymerMW \leq 21kDa$) emerged as the sole robust formulation driver, successfully replicating across all unseen drug subsets.
- **Fuzzy Boundary Robustness:** Translating static crisp thresholds into continuous *Low*, *Medium*, *High* membership functions effectively mitigated boundary rigidity. This representation reduced the Mean Absolute Error (MAE) by 0.1262 for near-threshold formulations while fully preserving global predictive accuracy across the entire dataset.

V. CONCLUSION

This work demonstrates that integrating fuzzy logic representations into baseline decision tree architectures addresses the critical trade-off between interpretability and boundary flexibility in pharmaceutical machine learning pipelines. By replacing rigid cutoff thresholds with continuous membership functions derived from leakage-free quantiles, the system eliminates decision instability for near-threshold formulations without sacrificing predictive performance.

Future developments will focus on expanding the fuzzy rule-base to capture multi-variable kinetic interactions and integrating automated membership optimization to further refine release profile forecasting for novel drug delivery systems.

REFERENCES

- [1] A. Goren, Z. Bao, J. P. Martinez Lozano, and C. Allen, "A formulation dataset of poly(lactide-co-glycolide) nanoparticles for small molecule delivery," *Scientific Data*, 2024.
- [2] Y. Bao *et al.*, "Machine learning benchmarks for plga microparticle drug release kinetics," *Journal of Controlled Release*, 2025.